Using the PegasusAI Bot in JupyterLab on Jetstream2 (Course Demo)

Quick mental model:

- **Your laptop = the window (web browser)**
- **Jetstream2 = the computer (remote VM in the cloud)**
- **PegasusAI bot = your assistant (inside JupyterLab)**

What this is (and why we use it)

- **Jetstream2** is a national cloud platform (ACCESS) where you run a virtual machine (VM).
- **JupyterLab** is a web-based IDE for running Python notebooks.
- **PegasusAI bot** is an AI assistant inside JupyterLab that helps you understand errors and improve code.

Important: The bot is a learning tool. It gives hints and explanations, but you're still responsible for solving the assignment.

What you need (prerequisites)

- An **ACCESS** account
- A **GitHub** account
- The **course Jetstream2 image** (provided in Lecture 1)
- A **GitHub Classroom invite link** for each assignment (given when the assignment is released)

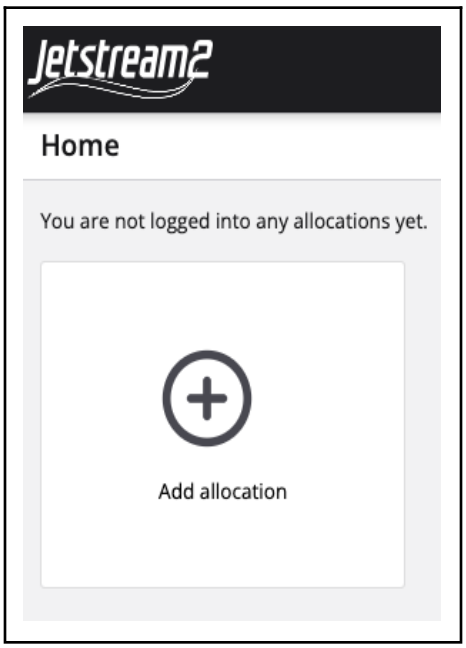
What you'll be able to do (takeaways)

- Launch your own Jetstream2 VM from a pre-built course image
- Open JupyterLab in your browser (running remotely on your VM)
- Use PegasusAI to help debug and learn
- Clone, run, and complete GitHub Classroom assignments on Jetstream2

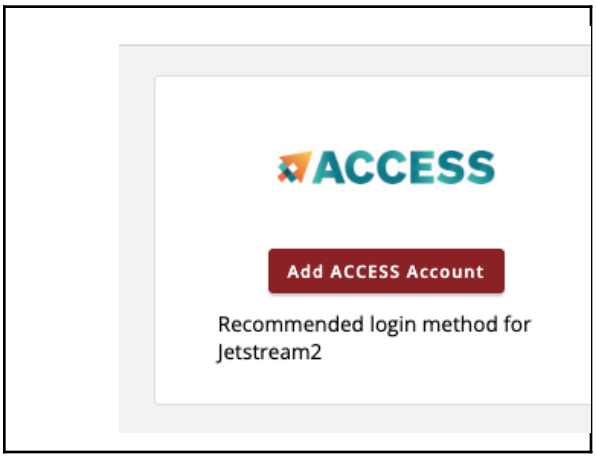
Part A — Do these steps ONE TIME (setup)

A1) Log in to Jetstream2 Exosphere

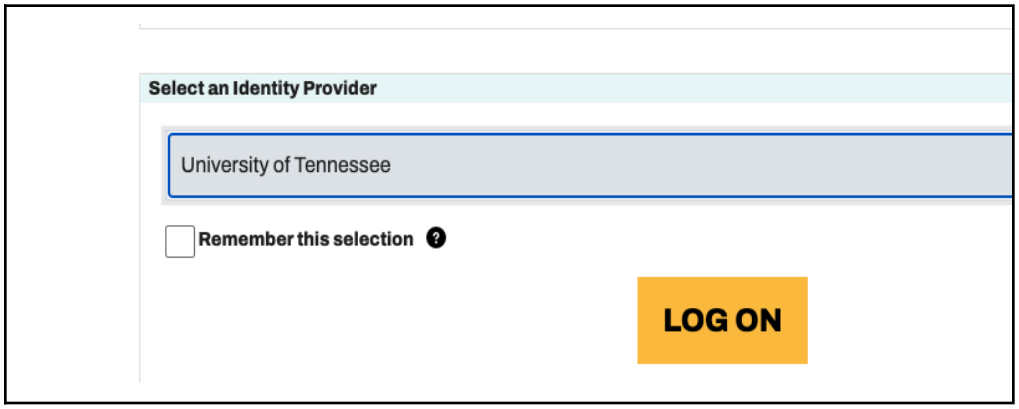
1. Open: `https://jetstream2.exosphere.app`
2. Click **Add Allocation**



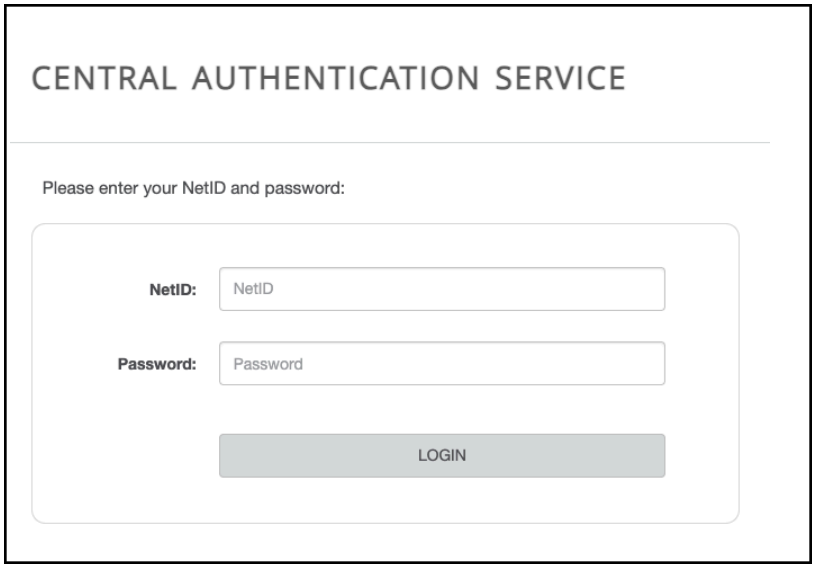
3. Click **Add ACCESS account**



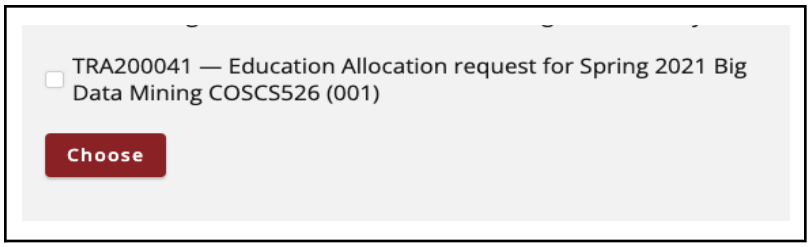
4. Identity provider: (e.g., **University of Tennessee)**, then click **LOG ON**



5. Log in with your **UTK NetID + password**

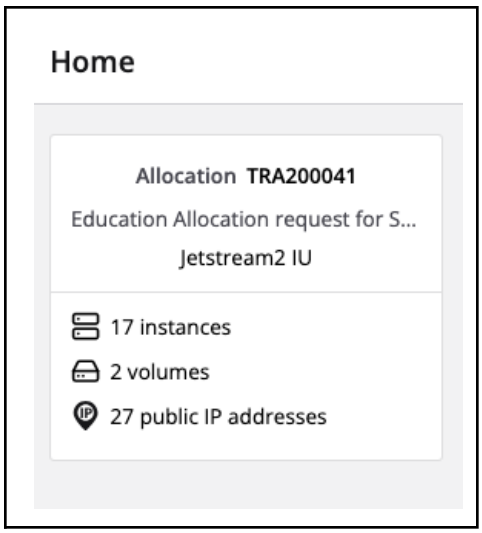


6. Select allocation **TRA200041**, then click **Choose**

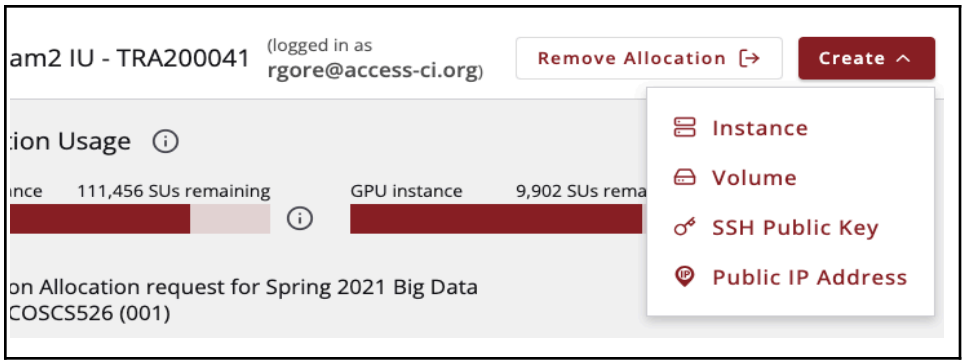


A2) Create your VM instance from the course image

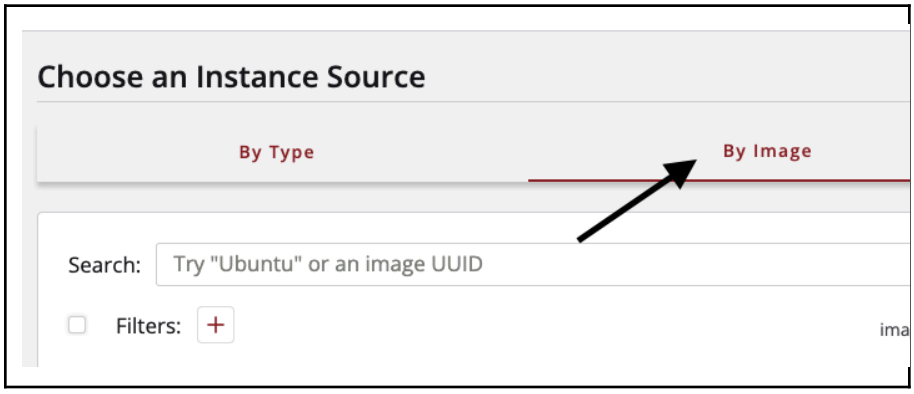
1. Click allocation **TRA200041**



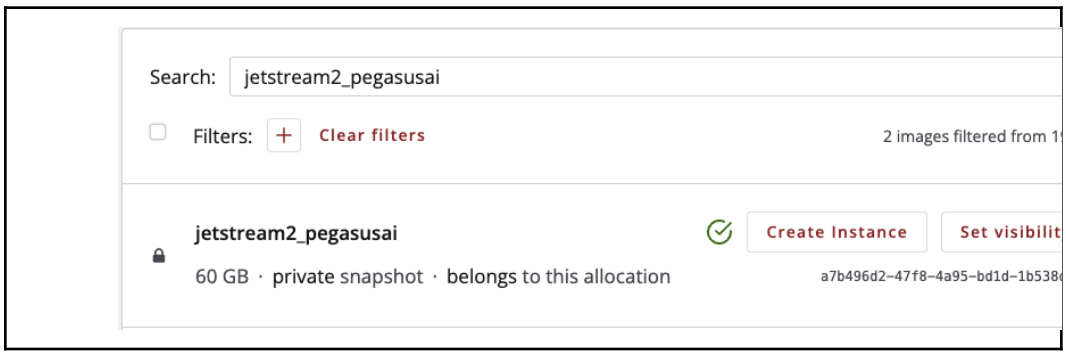
2. Click **Create** → choose **Instance**



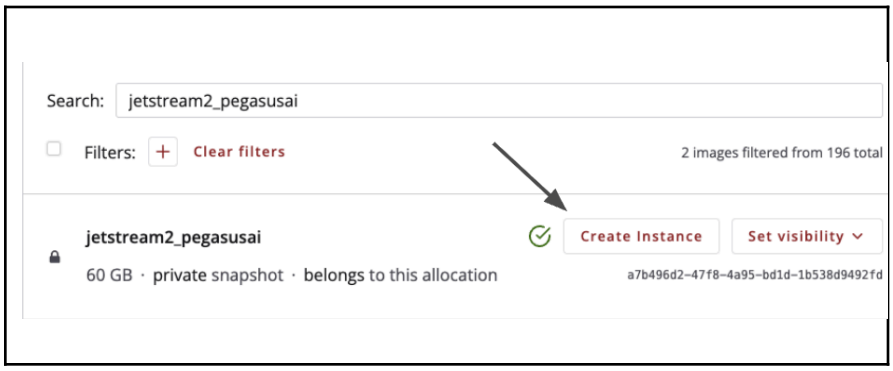
3. Choose the **By Image** tab



4. Search for image name:
   `jetstream2_pegasusai`



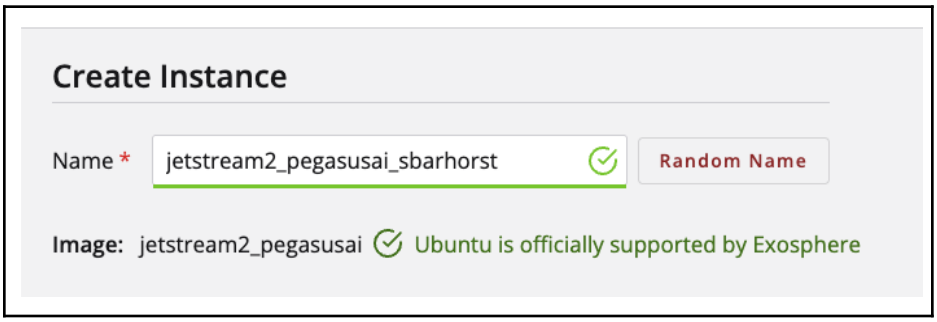
5. Click **Create Instance**



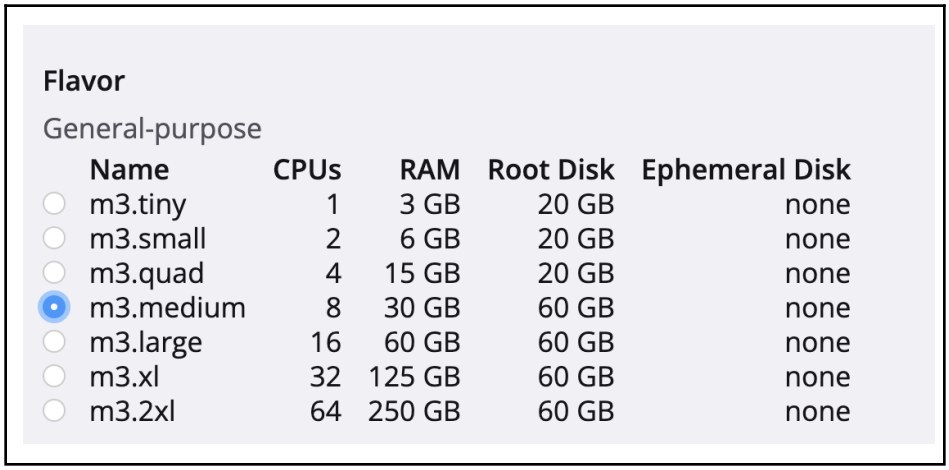
A3) Configure the instance

Use these settings:

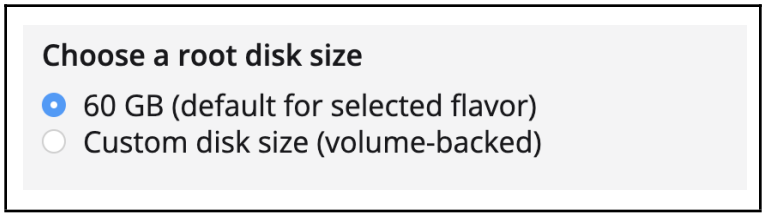
- **Instance name:** `jetstream2_pegasusai_<your_ID>`



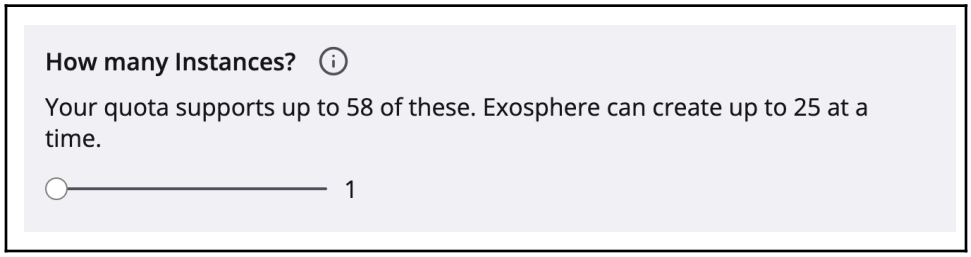
- **Flavor:** `m3.Medium` (General-purpose)



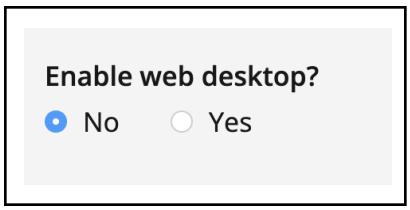
- **Root disk size:** Default for selected flavor



- **How many instances:** 1

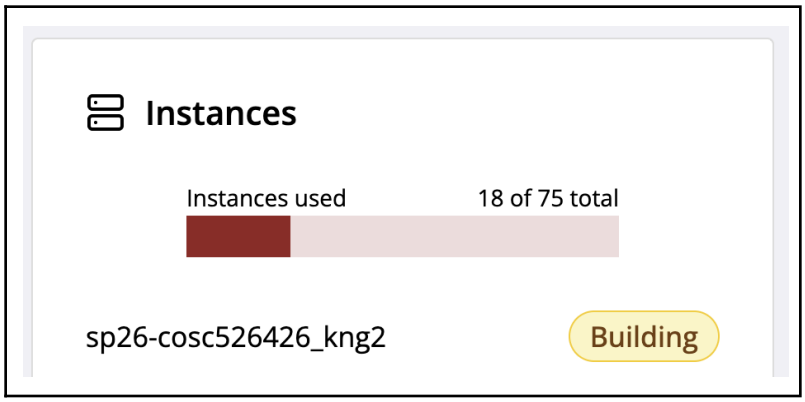


- **Web Desktop:** No



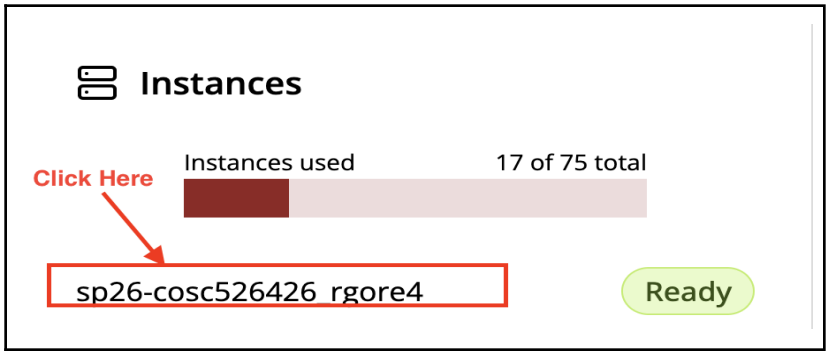
- **Do not change other settings**
- Click **Create** (bottom of page)

Wait until the status becomes **Ready** (build can take a few minutes).

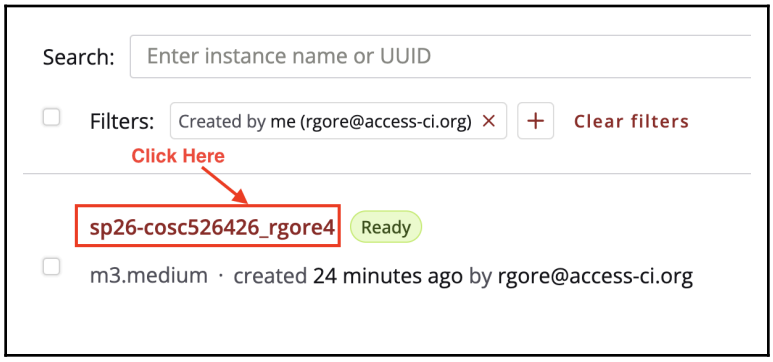


A4) Open the Web Shell (terminal)

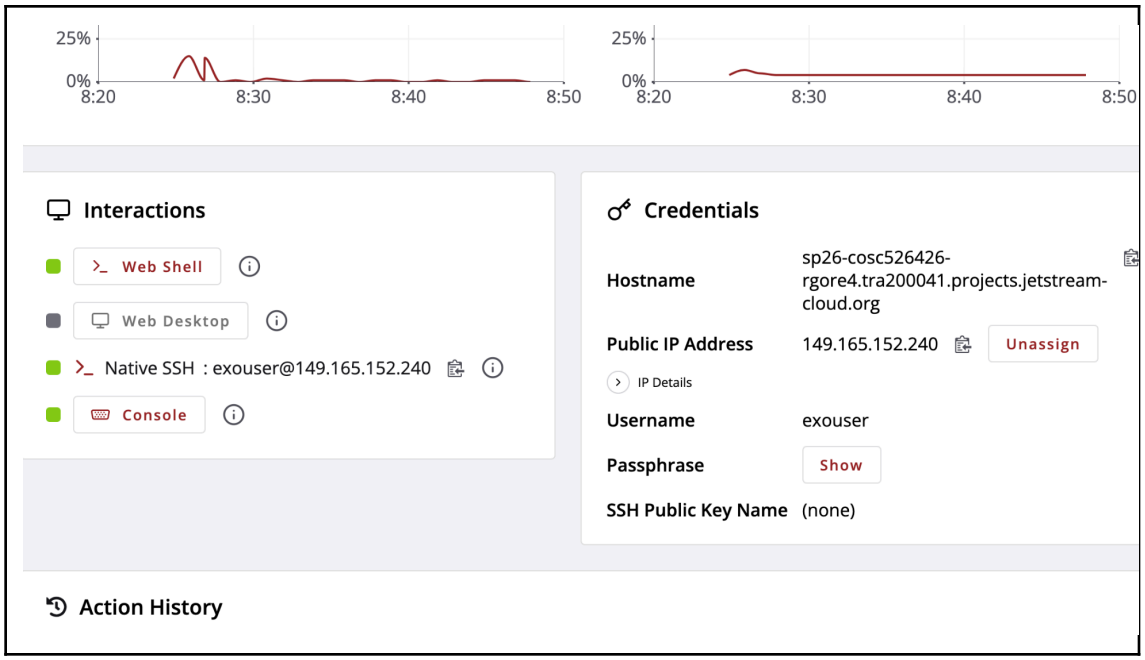
1. When status is **Ready**, click your **instance name**



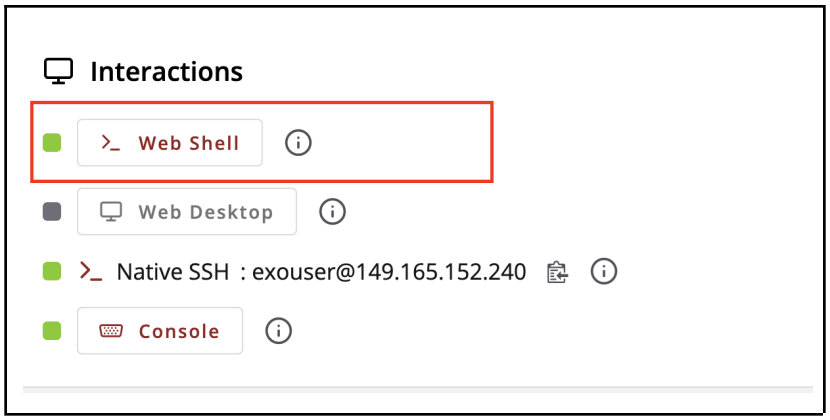
2. Click your **instance name again** in the list/details view



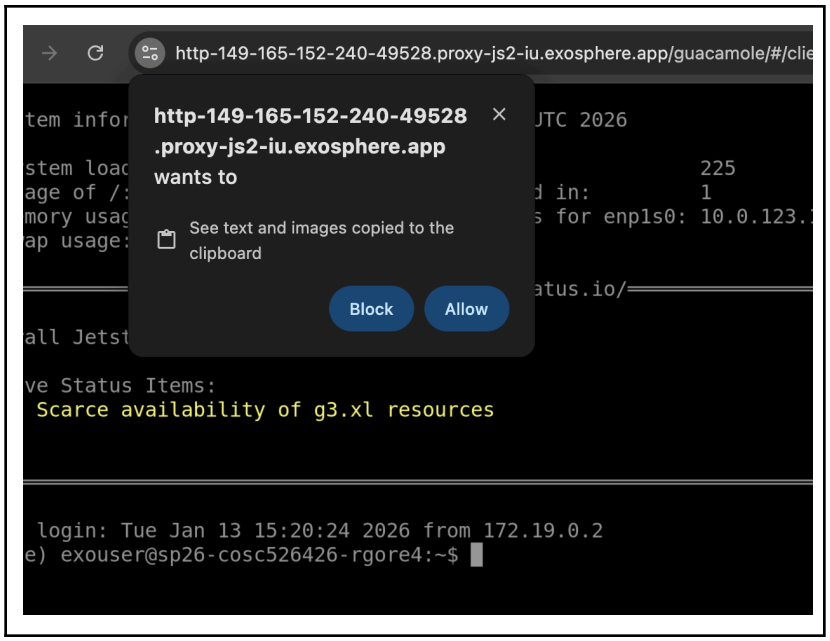
3. Confirm you can see instance details (hostname, IP, etc.)



4. Click the red **>_ Web Shell** button



5. A terminal opens in a new tab → click **Allow** if prompted



Part B — Do this ONE TIME (GitHub CLI login)

B1) Start GitHub authentication in the Web Shell

In the Web Shell terminal:

Run: `gh auth login`

1. Choose:

- **GitHub.com**

```
(base) exouser@sp26-cosc526426-rgore4:~$ gh auth login
? What account do you want to log into?  [Use arrows to move, type to
> GitHub.com
  GitHub Enterprise Server
```

- **HTTPS**

```
? What is your preferred protocol for Git operations on this host?  [Use arrows t
type to filter]
> HTTPS
  SSH
```

- **Login with a web browser**

```
? How would you like to authenticate GitHub CLI?  [Use arrows to move, type to f
> Login with a web browser
  Paste an authentication token
```

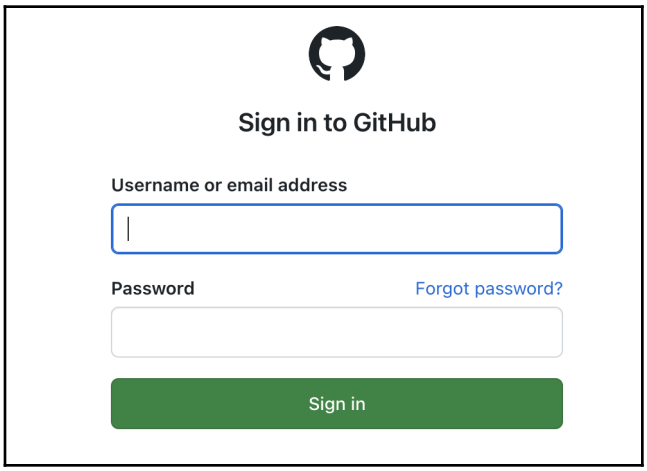
2. You'll see a **one-time code** in the terminal — **keep it handy**.

```
! First copy your one-time code: 0532-4B7C
Press Enter to open github.com in your browser...
```

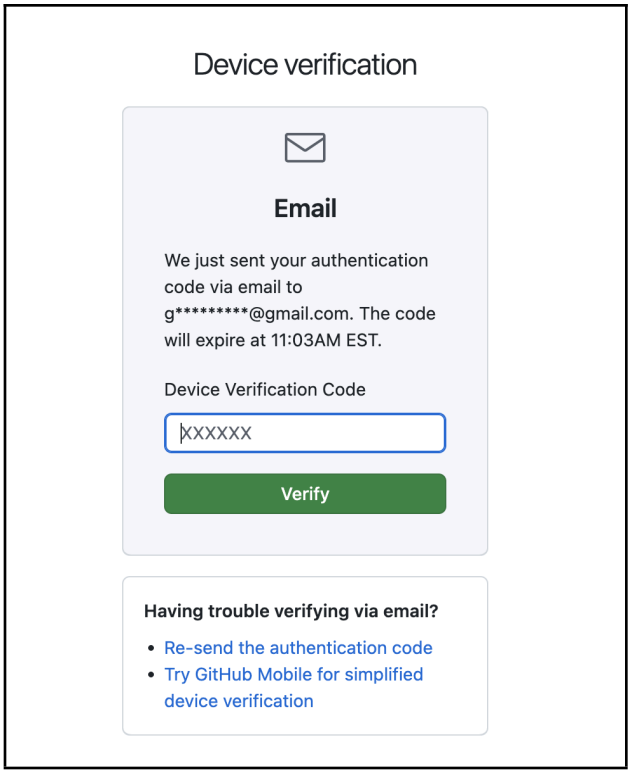
Note: The Web Shell usually cannot open a browser automatically. That's expected.

B2) Finish authentication in your browser

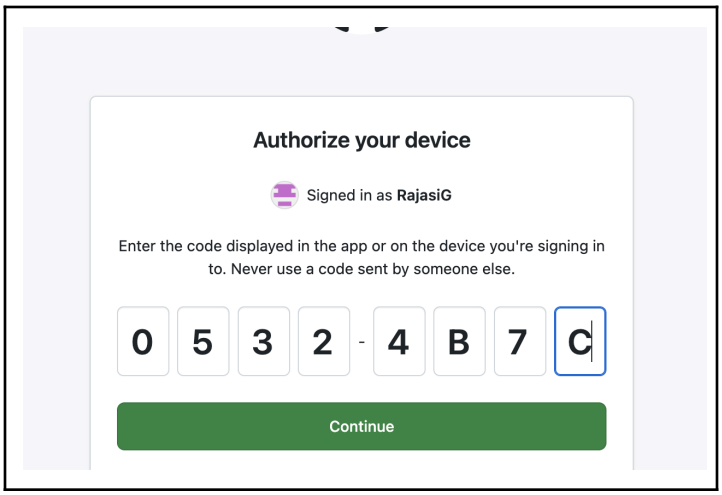
1. On your laptop browser, open: `https://github.com/login/device`
2. Sign in to GitHub



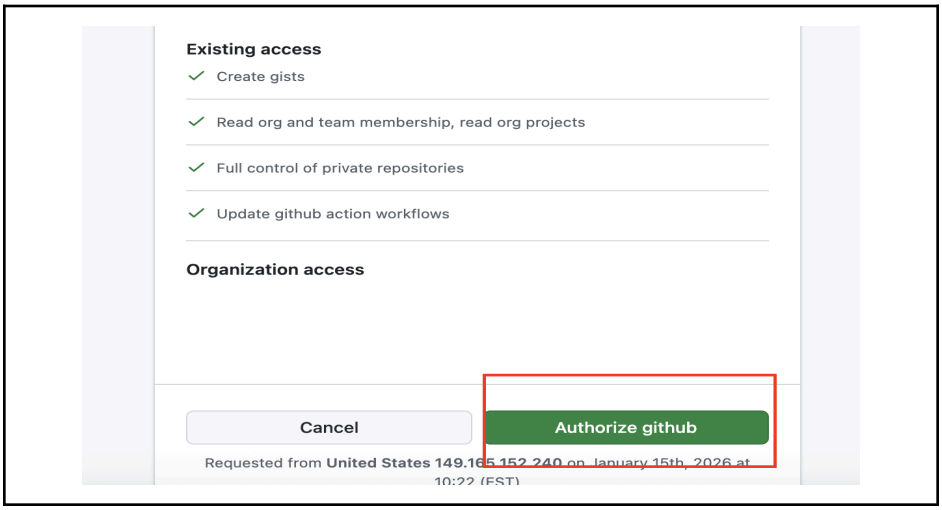
3. Follow GitHub's steps (you may need email or mobile verification)



4. When prompted, paste the **one-time code** shown in your Web Shell



5. Click **Authorize github**



6. Complete any final email verification prompts

When done, your GitHub CLI is authorized.

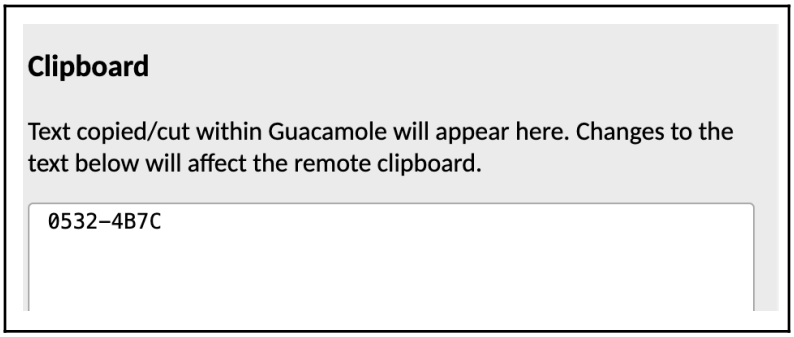
Copy/paste help (Web Shell → outside)

Sometimes copying from the Web Shell needs the Guacamole clipboard:

Mac

- Highlight the code in the Web Shell

- Press **Ctrl + Cmd + Shift** to open clipboard (code is copied)

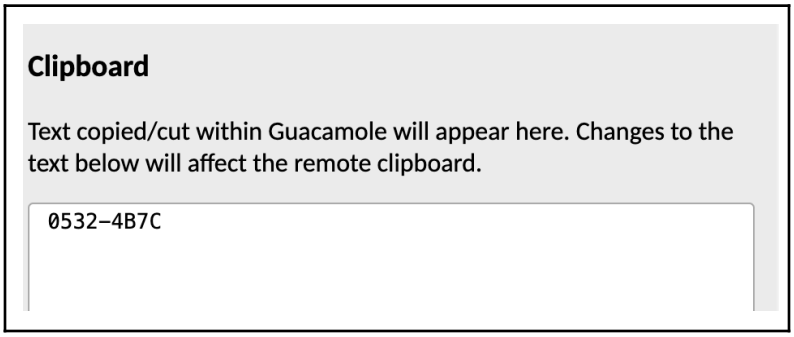


- Press **Ctrl + Cmd + Shift** to close clipboard
- Paste with **Ctrl + Shift + V**

Windows/Linux

- Highlight the code in the Web Shell

- Press **Ctrl + Alt + Shift** to open clipboard



- Press **Ctrl + Alt + Shift** to close clipboard
- Paste with **Ctrl + Shift + V** (or right-click paste)

Part C — Do these steps EVERY ASSIGNMENT (repeat)

C1) Clone the assignment repository (in Web Shell)

Before you start: A github repository link will be provided for you. Make sure that you have access and the repository exists.

1. In the Web Shell, run:

```
git clone https://github.com/CISC879-BigData/assignment03-demo.git
```

2. List folders to find your assignment directory:

```
ls
```

```
(base) exouser@sp26-cosc526426-rgore4:~$ ls
assignment03-pegasusai-RajasiG
```

3. Enter the assignment folder:

```
cd <REPO_FOLDER_NAME>
```

```
(base) exouser@sp26-cosc526426-rgore4:~$ cd assignment03-pegasusai-Rajas
(base) exouser@sp26-cosc526426-rgore4:~/assignment03-pegasusai-RajasiG$
```

C2) Start JupyterLab (in Web Shell)

1. Load the environment:

   ```
   module load miniforge
   ```

   (No output usually means it worked.)

   ```
   (base) exouser@sp26-cosc526426-rgore4:~/assignment03-pegasusai-RajasiG$ module load miniforge
   (base) exouser@sp26-cosc526426-rgore4:~/assignment03-pegasusai-RajasiG$
   ```

2. Start Jupyter:

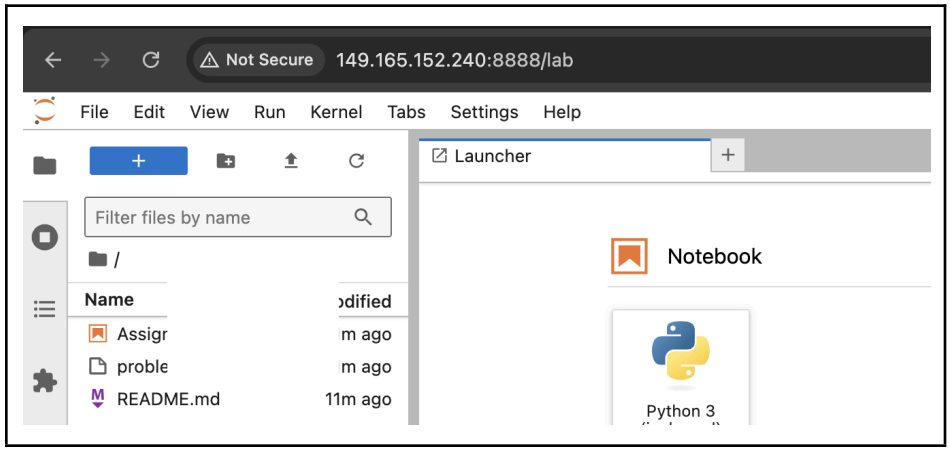
   ```
   jupyter-ip.sh
   ```

   ```
   (base) exouser@sp26-cosc526426-rgore4:~/assignment03-pegasusai-RajasiG$ jupyter-ip.sh
   [I 2026-01-15 16:29:35.895 ServerApp] jupyter_lsp | extension was successfully linked.
   [I 2026-01-15 16:29:35.898 ServerApp] jupyter_server_terminals | extension was successfully linked.
   ```

3. Copy the **second URL** shown (the one with your instance IP address)

4. Paste it into a **new browser tab** on your laptop

5. In JupyterLab, you'll see the jupyter notebook in the left panel

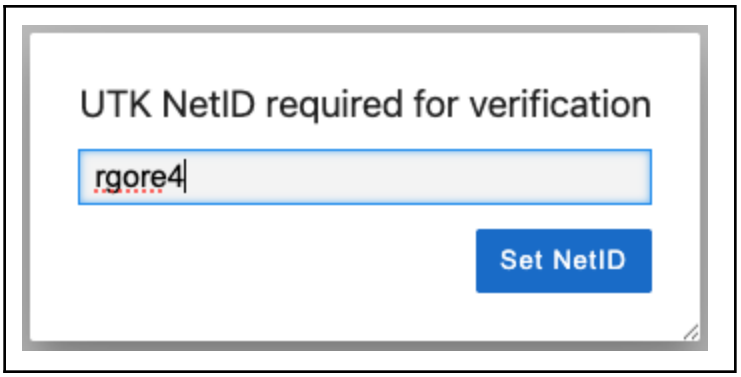


C3) Use the PegasusAI bot in JupyterLab (Homework 3 and 4 ONLY - Skip this step otherwise)

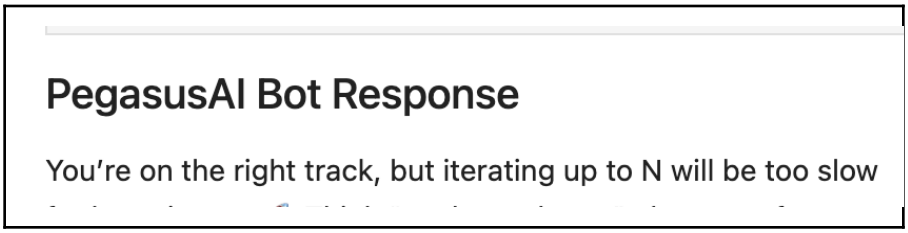
1. Run a notebook cell

2. If there's an error (or you want help), click the **(?) icon** on the right side of the cell toolbar

3. You will get a first-time only pop up to enter your UTK ID. Here you can enter any string (e.g., user)



4. The bot will generate suggestions based on that cell's output



Part D - Steps to shelve Jetstream2 instances (Repeat for every assignment)

D1: Shutting down Jupyter

1. Close the JupyterLab browser tab
   (This just closes the interface in your browser. The server and kernels are still running in the background.)

2. Go to the web shell where Jupyter is running

3. Press `Ctrl + C` twice

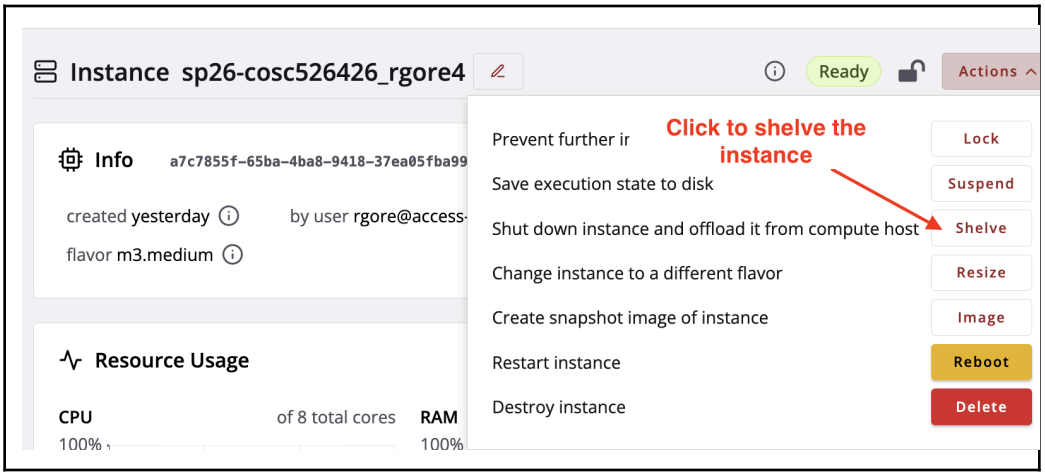
4. Close the web shell

D2: Shelving your instance from the Jetstream2 website

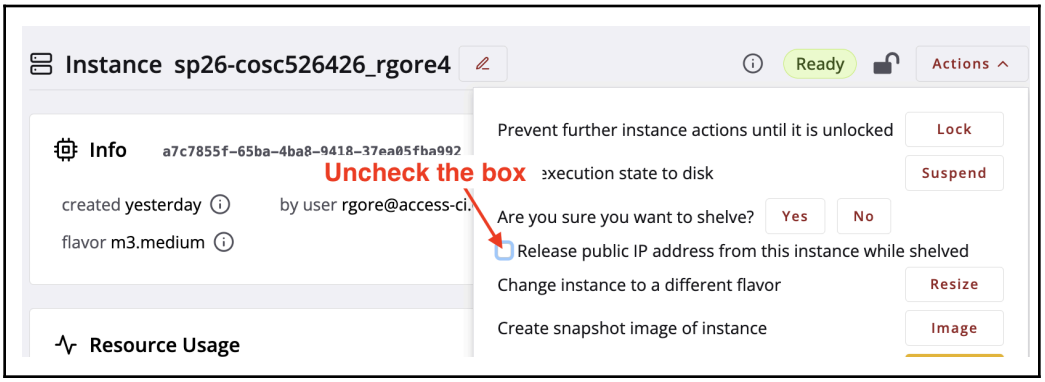
1. Go back to your Jetstream instance

2. In the **top-right corner**, click the **Action** drop-down button

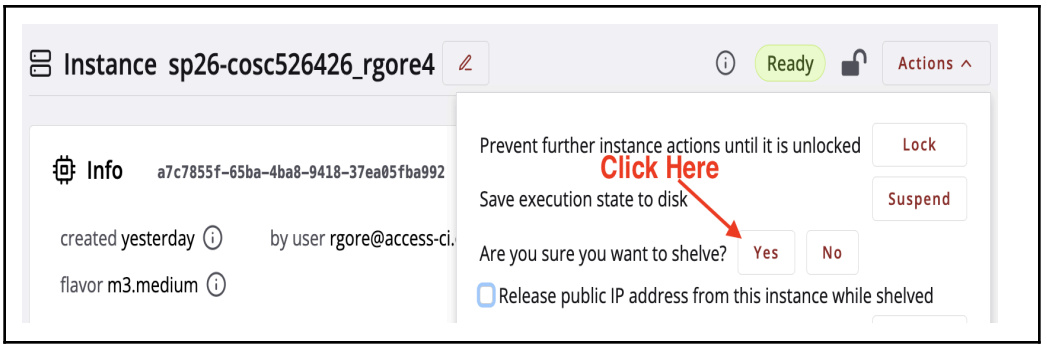
3. Select **Shelve**, and **do not click "Yes"** in the confirmation prompt until you have read the next instruction.



4. **Very crucial step: Uncheck** the box labeled **Release IP address**
   a. If you do not do this step, you will have to repeat Part B of these instructions the next time you start your instance



5. Click **Yes**



Tip for smooth workflow

- Keep **two tabs** open:
    1. **Web Shell** (terminal)
    2. **JupyterLab** (notebooks)

That's it — once setup is done, each new assignment is basically: **clone → start Jupyter → work with bot support**.